

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T09 · Rounding, Estimation and Bounds

With Answers

Questions with full worked solutions

■ 48 pages

FAMILY MAP

Rounding Estimation Bounds

F01	Recover single-measurement bounds	7 Q 12 marks
F02	Optimise a compound expression using bounds	11 Q 35 marks
F03	Round to a stated accuracy	2 Q 10 marks
F04	Estimate a numerical calculation	2 Q 6 marks

Working Unit 01

4MA1-1H Q2

ORIGINAL QUESTION**2 marks**

The height of each cup is 9 cm, correct to the nearest cm

(b) Write down the lower bound of the height.

..... cm
(1)

The weight of each cup is 120 g, correct to the nearest 10 g

(c) Write down the upper bound of the weight.

..... g
(1)

Worked Solution 01

Bounds from stated accuracy

UNIT 1
2 marks**Height**

$$9 \text{ cm to the nearest cm} \Rightarrow 8.5 \leq h < 9.5.$$

Weight

$$120 \text{ g to the nearest 10 g} \Rightarrow 115 \leq w < 125.$$

Final answer

$$(b) \ 8.5 \text{ cm} \quad (c) \ 125 \text{ g}$$

Dependency check Independent of the excluded Unit 2 profit part Q2(a).

Working Unit 24

4MA1-2H Q22

ORIGINAL QUESTION

4 marks

- 22 The diagram shows a square inside rectangle $ABCD$

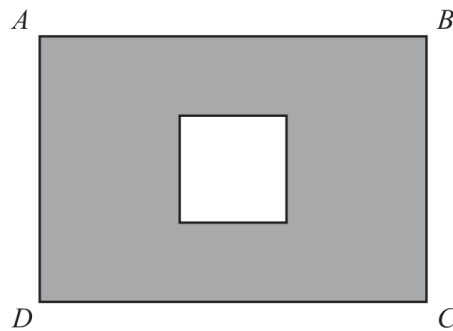


Diagram **NOT**
accurately drawn

The total area of the region shown shaded in the diagram is $X\text{cm}^2$

$AB = 11.5\text{ cm}$ correct to the nearest 0.5 cm

$BC = 9.2\text{ cm}$ correct to 2 significant figures

side of square $= 4.1\text{ cm}$ correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X =$

(Total for Question 22 is 4 marks)

Worked Solution 24

Bounds for a shaded area

UNIT 1
4 marks**Input bounds**

$$11.25 \leq AB < 11.75, \quad 9.15 \leq BC < 9.25, \quad 4.05 \leq s < 4.15.$$

Upper shaded area

$$X_U = 11.75(9.25) - 4.05^2 = 92.285.$$

Lower shaded area

$$X_L = 11.25(9.15) - 4.15^2 = 85.715.$$

Choose the safe accuracy

$$85.715 \leq X < 92.285; \text{ both bounds round to 90 to 1 significant figure.}$$

Final answer

$$X = 90 \text{ cm}^2 \text{ (1 s.f.)}$$

Question 26

4MA1-2H Q19

ORIGINAL QUESTION

3 marks

19 $G = \frac{c}{2f - 3h}$

$c = 8$ correct to the nearest whole number

$f = 6.62$ correct to 2 decimal places

$h = 1.2$ correct to 1 decimal place

Work out the lower bound for the value of G

Give your answer correct to 3 decimal places.

Show your working clearly.

Worked Solution 26

Lower bound

MODULAR UNIT 1

3 marks

Bounds

$$c_L = 7.5, f_U = 6.625, h_L = 1.15.$$

Substitute

$$G_L = \frac{7.5}{2(6.625) - 3(1.15)} = 0.765306 \dots$$

Final answer

$$0.765$$

Question 29

4MA1-2H Q17 whole

ORIGINAL QUESTION UNIT 1

3 marks

17 $P = a(c + y)$

 $a = 8.3$ correct to 2 significant figures $c = 2$ correct to 1 significant figure $y = 15$ correct to the nearest 5Work out the upper bound for the value of P

Show your working clearly.

.....

Worked solution 29

Upper bound of a product with a sum

MODULAR UNIT 1

3 marks

whole – Upper bound of a product with a sum

Choose upper bounds

8.3 to 2 s.f. gives $8.25 \leq a < 8.35$; 2 to 1 s.f. gives $1.5 \leq c < 2.5$; 15 to nearest 5 gives $12.5 \leq y < 17.5$.

Use the largest permitted values

$$P_{UB} = 8.35(2.5 + 17.5).$$

Evaluate

$$P_{UB} = 8.35(20) = 167.$$

Final answer

167

Question 29

4MA1-2H Q25

ORIGINAL QUESTION

4 marks

- 25 A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.
- Work out the upper bound for the density of the sphere.
Give your answer in g/cm^3 correct to 2 decimal places.
Show your working clearly.

..... g/cm^3

Worked Solution 29

Upper bound for sphere density

MODULAR UNIT 1

4 marks

Choose the worst-case bounds

$$r_{LB} = 2.75 \text{ cm}, \quad M_{UB} = 265\pi \text{ g}.$$

Use the lower volume

$$V_{LB} = \frac{4}{3}\pi(2.75)^3.$$

Maximise density

$$\rho_{UB} = \frac{265\pi}{(4/3)\pi(2.75)^3} = 9.56 \text{ g/cm}^3 \text{ (2 d.p.)}.$$

Final answer

$$9.56 \text{ g/cm}^3$$

Original question 16

4MA1-1H Q21 M01

MODULAR UNIT 1

3 marks

21 $T = \frac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.

Show your working clearly.

.....

Worked solution 16

Use upper and lower bounds

MODULAR UNIT 1

3 marks

M01 – Use upper and lower bounds**Write the bounds**

$$UB(x) = 28.45, UB(y) = 17.5, LB(w) = 87.5.$$

Substitute

$$UB(T) = \frac{28.45^2 + 17.5^2}{87.5} = 12.7503\dots$$

Round

$$UB(T) = 12.8 \text{ (3 s.f.)}.$$

Final answer

12.8

Original question 18

4MA1-2H Q2 M01

MODULAR UNIT 1

1 marks

2 The length of a table is measured as 1.4 metres correct to one decimal place.

(a) Write down the upper bound of the length of the table.

..... metres

(1)

Worked solution 18

Upper bound of a measured length

MODULAR UNIT 1

1 marks

M01 – Upper bound of a measured length**Use half the rounding unit**1.4 m correct to 1 decimal place gives $1.35 \leq L < 1.45$.**Read the upper bound**

$$UB = 1.45 \text{ m.}$$

Final answer

1.45 m

Original question 19

4MA1-2H Q2 M02

MODULAR UNIT 1

1 marks

2 The length of a table is measured as 1.4 metres correct to one decimal place.

(b) Write down the lower bound of the length of the table.

..... metres

(1)

Worked solution 19

Lower bound of a measured length

MODULAR UNIT 1

1 marks

M02 – Lower bound of a measured length**Use half the rounding unit**

$$1.35 \leq L < 1.45.$$

Read the lower bound

$$LB = 1.35 \text{ m.}$$

Final answer

1.35 m

Original question 25

4MA1-2H Q18 Q18

ORIGINAL QUESTION**3 marks**

18 $X = \frac{2a - b}{f}$

$a = 7.5$ correct to 1 decimal place.

$b = 3.42$ correct to 2 decimal places.

$f = 2$ correct to the nearest whole number.

Work out the upper bound of the value of X

Show your working clearly.

Worked solution 25

Chapter: Rounding, Estimation & Bounds

MODULAR UNIT 1

3 marks

Q18 – Chapter: Rounding, Estimation & Bounds

Family: Optimise a compound expression using bounds • Variant: Algebraic expression upper or lower bound

Method

The relevant bounds are $a < 7.55$, $b \geq 3.415$, and $f \geq 1.5$.

Why: Use the stated rounding accuracies.

Method

Because the numerator and denominator are positive, the upper bound is $(2 \times 7.55 - 3.415)/1.5$.

Why: Maximise the numerator and minimise the positive denominator.

Method

The upper bound is $11.685/1.5 = 7.79$.

Why: Evaluate the extremal expression exactly to the terminating decimal.

Final answer

upper bound: 7.79

Original question 15

4MA1-1H Q18

ORIGINAL QUESTION**3 marks**

18 $A = w - \frac{x^2}{y}$

$w = 3.45$ correct to 2 decimal places.

$x = 1.9$ correct to 1 decimal place.

$y = 5$ correct to the nearest whole number.

Work out the lower bound of the value of A
Show your working clearly.

Worked solution 15

Chapter: Rounding, Estimation & Bounds

MODULAR UNIT 1

3 marks

Q18 – Chapter: Rounding, Estimation & Bounds

Family: Optimise a compound expression using bounds • Variant: Algebraic expression upper or lower bound

Method $w \geq 3.445$, $x < 1.95$ and $y \geq 4.5$.*Why: Use the correct bounds for the stated rounding.***Method**For a lower bound of $A = w - x^2/y$, use $w = 3.445$, $x = 1.95$ and $y = 4.5$.*Why: Minimise w and maximise x^2/y .***Method** $A = 3.445 - (1.95)^2/4.5 = 2.6$ (approximately).*Why: Substitute the bounds and calculate.***Final answer**

lower bound of A: 2.6

Original question 21

4MA1-2H Q4 Q4(a)

ORIGINAL QUESTION**1 marks**

- 4 Each side of a regular octagon has a length of 18 mm, correct to the nearest 0.5 mm

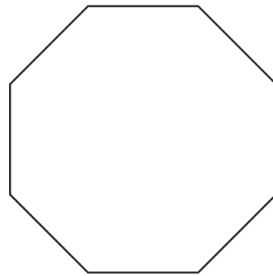


Diagram **NOT**
accurately drawn

- (a) Write down the lower bound of the length of each side of the octagon.

..... mm
(1)

Worked solution 21

Chapter: Rounding, Estimation & Bounds

MODULAR UNIT 1

1 marks

Q4(a) – Chapter: Rounding, Estimation & Bounds

Family: Recover single-measurement bounds • Variant: Direct lower and upper bounds

Method18 mm correct to the nearest 0.5 mm has lower bound $18 - 0.25 = 17.75$ mm.*Why: Half the rounding interval is 0.25 mm.***Final answer****lower bound:** 17.75 mm

Original question 22

4MA1-2H Q4 Q4(b)

ORIGINAL QUESTION

1 marks

(b) Write down the upper bound of the length of each side of the octagon.

..... mm

(1)

Worked solution 22

Chapter: Rounding, Estimation & Bounds

MODULAR UNIT 1

1 marks

Q4(b) – Chapter: Rounding, Estimation & Bounds

Family: Recover single-measurement bounds • Variant: Direct lower and upper bounds

MethodThe upper bound is $18 + 0.25 = 18.25$ mm.*Why: Add half the rounding interval.***Final answer****upper bound:** 18.25 mm

16 $P = \frac{2a - c}{d}$

$a = 58.4$ correct to 3 significant figures.

$c = 20$ correct to 2 significant figures.

$d = 3.6$ correct to 2 significant figures.

Work out the upper bound for the value of P .

Show your working clearly.

Give your answer correct to 2 decimal places.

Worked solution

January 2020 | Question 16 | 3 marks

Family: Optimise a compound expression using bounds • Variant: Algebraic expression upper or lower bound

Method / working

1. Upper bounds: $a < 58.45$, $c \geq 19.5$, $d \geq 3.55$.
 2. $P = (2a - c)/d$ is maximised by largest numerator and smallest denominator.
 3. $P < [2(58.45) - 19.5]/3.55 = 27.4366\dots$, so upper bound = 27.44.
- Final answer: 27.44

15

$$a = \frac{v - u}{t}$$

$v = 9.6$ correct to 1 decimal place

$u = 3.8$ correct to 1 decimal place

$t = 1.84$ correct to 2 decimal places

Calculate the upper bound for the value of a .

Give your answer as a decimal correct to 2 decimal places.

Show your working clearly.

Worked solution

January 2021 | Question 15 | 3 marks

Family: Optimise a compound expression using bounds • Variant: Algebraic expression upper or lower bound

Method / working

1. Upper bounds: $v=9.65$, lower bounds: $u=3.75$ and $t=1.835$.

2. Use $a < 9.65 / (3.75 \times 1.835) = 3.2152\dots$

3. To 2 d.p., the upper bound is 3.22.

Final answer: 3.22

20 $P = \frac{t - w}{y}$

$t = 9.7$ correct to 1 decimal place

$w = 5.9$ correct to 1 decimal place

$y = 3$ correct to 1 significant figure

Calculate the upper bound for the value of P .
Show your working clearly.

Worked solution

June 2021 | Question 20 | 3 marks

Family: Optimise a compound expression using bounds • Variant: Algebraic expression upper or lower bound

Method / working

1. Use bounds $t < 9.7$, $w = 5.9$, $y = 3$: $UB(t) = 9.75$, $LB(w) = 5.85$, $LB(y) = 2.5$.
2. Compute $(9.75 - 5.85) / 2.5 = 1.56$.
3. State the upper bound.

Final answer: 1.56

13 Here is a triangle XYZ .

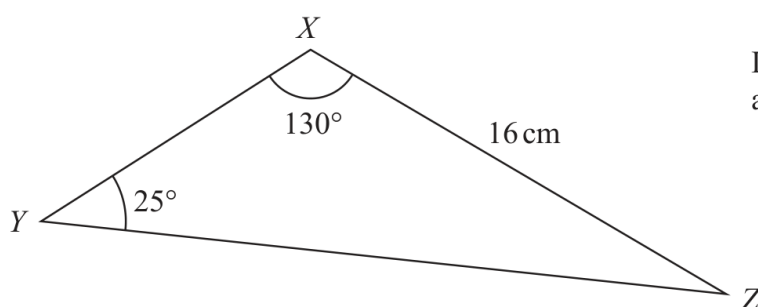


Diagram **NOT**
accurately drawn

The length XZ and the angles YXZ and XYZ are each given correct to 2 significant figures.

Calculate the upper bound for the length YZ .

Give your answer correct to one decimal place.

Show your working clearly.

..... cm

Worked solution

November 2021 | Question 13 | 3 marks

Family: Optimise a compound expression using bounds | Variant: Geometry, rate or practical decision bounds

METHOD / WORKING

1. Using the sine rule,

$$2. \frac{YZ}{\sin X} = \frac{XZ}{\sin Y}$$

3. For the upper bound of YZ :

$$4. XZ < 16.5, \quad X < 25.5^\circ, \quad Y < 135^\circ$$

$$5. YZ_{\text{upper}} = \frac{16.5 \sin 25.5^\circ}{\sin 135^\circ} = 10.0457 \dots$$

6. Correct to one decimal place,

$$7. 10.0457 \dots \approx 10.0$$

8. Answer: 10.0 cm

FINAL ANSWER

10.0 cm

- 17 A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3}) \text{ m}^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in kg/m^3 correct to 1 decimal place.
Show your working clearly.

..... kg/m^3

(Total for Question 17 is 4 marks)

Answer reference | May-Nov 2020 | Question 17

Family: Recover single-measurement bounds • Variant: Mixed stated accuracies

17	5025 or 5.025 or 4975 or 4.975		4	B1 Accept 5024.9 for 5025 or 5.0249 for 5.025
	1.845×10^{-3} oe or 1.835×10^{-3} oe			B1 Accept 1.8449×10^{-3} for 1.845×10^{-3}
	$\frac{5.025}{1.835 \times 10^{-3}} (= 2738.4...) \text{ oe}$			M1 for correct substitution into $\frac{m_{UB}}{v_{LB}}$ where $5 < m_{UB} \leq 5.025$ and $1.835 \times 10^{-3} \leq v_{LB} < 1.84 \times 10^{-3}$
		2738.4		A1 dep on correct working
				Total 4 marks

Worked solution

May-Nov 2020 | Question 17 | 4 marks

Family: Recover single-measurement bounds | Variant: Mixed stated accuracies

METHOD / WORKING

Worked solution
May-Nov 2020 Question 17 4 marks
Family: Recover single-measurement bounds Variant: Mixed stated accuracies
Method
Density is
$density = \frac{mass}{volume}$
To get the upper bound for density, use the upper bound for mass and the lower bound for volume.
The mass is 5 kg correct to the nearest 50 g.
50 g=0.05 kg
Half of this is 0.025 kg, so
upper bound for mass=5+0.025=5.025
The volume is $1.84 \times 10^{-3} = 0.00184 m^3$, correct to 3 significant figures.
The lower bound is
$0.00184 - 0.000005 = 0.001835$
So the upper bound for the density is
$(5.025) / (0.001835) = 2738.419 \dots$
Correct to 1 decimal place,
2738.4
Answer: $2738.4 kg/m^3$

FINAL ANSWER

19 The acceleration, a , of an object is given by

$$a = \frac{v - u}{t}$$

where

$v = 45.23$ correct to 2 decimal places

$u = 5.12$ correct to 2 decimal places

$t = 8.5$ correct to 2 significant figures

By considering bounds, work out the value of a to a suitable degree of accuracy.
Show your working clearly and give a reason for your answer.

$a = \dots\dots\dots$

(Total for Question 19 is 5 marks)

Answer reference | January 2023 | Question 19

Family: Round to a stated accuracy • Variant: Round to decimal places or significant figures

19	45.225 or 45.235 or 5.115 or 5.125 or 8.45 or 8.55		5	B2 for all 6 correct (B1 for 4 or 5 correct) Accept $45.234\dot{9}$ for 45.235 $5.124\dot{9}$ for 5.125 $8.54\dot{9}$ for 8.55
	$\frac{45.235 - 5.115}{8.45} (= 4.7479\dots)$			M1 for correct substitution into the <i>UB</i> $a = \frac{v-u}{t}$ where $45.23 < v_{(UB)} \leq 45.235$ $5.115 \leq u_{(LB)} < 5.12$ $8.45 \leq t_{(LB)} < 8.5$
	$\frac{45.225 - 5.125}{8.55} (= 4.6900\dots)$			M1 for correct substitution into the <i>LB</i> $a = \frac{v-u}{t}$ where $45.225 \leq v_{(LB)} < 45.23$ $5.12 < u_{(UB)} \leq 5.125$ $8.5 < t_{(UB)} \leq 8.55$
	Working required	4.7 and correct reason		A1 dep on M2 4.7 and both answers round to 4.7 oe e.g. 1 dp or 2 sf
Total 5 marks				

Worked solution

January 2023 | Question 19 | 5 marks

Family: Round to a stated accuracy • Variant: Round to decimal places or significant figures

Official final mark-scheme pages are embedded immediately after this question.

5 The weight of a cake is 2.75 kg, correct to 2 decimal places.

(a) Write down the lower bound of the weight of the cake.

..... kg
(1)

(b) Write down the upper bound of the weight of the cake.

..... kg
(1)

Penny has worked out $\frac{81.3 \times 59.2}{1.9^2}$ on her calculator.

Her answer is 13 332.299 17

Penny's answer is not sensible.

(c) By rounding each number to one significant figure, work out a suitable estimate to show that her answer is not sensible.
Show your working clearly.

(2)

(Total for Question 5 is 4 marks)

Answer reference | January 2023 | Question 5

Family: Estimate a numerical calculation • Variant: One-significant-figure estimate

Question	working	Answer	Mark	Notes
5 (a)		2.745	1	B1
(b)		2.755	1	B1 allow 2.7549
(c)	$(80 \times 60) \div 2^2$		2	M1 For two of 80, 60, 2 or 4 rather than 2^2 oe
	eg $(80 \times 60) \div 2^2 = 1200$ oe <i>working with rounded values seen required</i>	1200		A1 dep on M1 for answer coming from use of the 3 rounded numbers – if 1200 seen then ignore any other working and comments
				Total 4 marks

Worked solution

January 2023 | Question 5 | 4 marks

Family: Estimate a numerical calculation • Variant: One-significant-figure estimate

Official final mark-scheme pages are embedded immediately after this question.

- 2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

(Total for Question 2 is 2 marks)

Worked solution

November 2025 | Question 2 | 2 marks

Family: Estimate a numerical calculation • Variant: One-significant-figure estimate

Method

Round each number to one significant figure:

2.11approx 2,quad 58.9approx 60,quad 8.859approx 9

$(2.11^2 \times 58.9)/(\text{sqrt}(8.859))$

approx

$(2^2 \times 60)/(\text{sqrt } 9)$

$=(4 \times 60)/(3)=80$

Answer: 80

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

7 The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

..... g
(1)

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

..... cm
(1)

(Total for Question 7 is 2 marks)

Worked solution

November 2025 | Question 7 | 2 marks

Family: Recover single-measurement bounds • Variant: Mixed stated accuracies

Method

The weight 475 g correct to the nearest g has lower bound

$$475 - 0.5 = 474.5$$

The height 120 cm correct to the nearest 10 cm has upper bound

$$120 + 5 = 125$$

Answer: (a) 474.5 g, (b) 125 cm

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

24 $D = \frac{n}{p - q}$

$n = 10.3$ correct to 1 decimal place

$p = 7.24$ correct to 2 decimal places

$q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of D to a suitable degree of accuracy.
Show your working clearly.

(Total for Question 24 is 5 marks)

Worked solution

November 2025 | Question 24 | 5 marks

Family: Round to a stated accuracy • Variant: Round to decimal places or significant figures

Method

$$D = (n) / (p - q)$$

Bounds:

$$10.25 \leq n < 10.35$$

$$7.235 \leq p < 7.245$$

$$4.385 \leq q < 4.395$$

Lower bound:

$$D_{\text{(lower)}} = (10.25) / (7.245 - 4.385) = 3.5839 \dots$$

Upper bound:

$$D_{\text{(upper)}} = (10.35) / (7.235 - 4.395) = 3.6443 \dots$$

So

$$3.5839 \dots < D < 3.6443 \dots$$

Both bounds round to 3.6 to 1 decimal place.

Answer: $D = 3.6$ to a suitable degree of accuracy